

Maximum Determinants of 55×55 Circulant Matrices over $\{0, 1\}$ and $\{-1, 1\}$

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Abstract

We determine the maximum determinants of circulant matrices of order 55 over both binary alphabets:

$$D_{01}(55) = 134694094094758395331307111329132,$$

$$D_{\pm}^{\text{circ}}(55) = 297532404423965431213849494795385191907933028352.$$

The normalized sign maximum is 16516366298178510386365752382353. The binary maximizers have weight 28 and form one affine orbit of 2200 words. The sign maximizers have 23 or 32 plus signs and form two affine orbits of 2200 words each, paired by global negation. The exact binary-to-sign bridge shows why these objectives have different winners. The proofs combine spectral moments, capped product bounds, cyclotomic folds, correlation-profile enumeration and binary lifts. The sign screen retains 17,835 targets and 151,772 margin tasks. Production and independent lifts each inspect 113,997,335,580 joined words and agree on every task and every returned word. Independent profile generation, explicit negative-spectrum certificates, small-order full-domain regressions and three exact determinant calculations complete the audit. To the best of our knowledge neither exact order-55 maximum was previously published in the sources checked.

1 Introduction

Binary circulant matrices are a structured class of integer matrices whose determinants can be evaluated spectrally, while the underlying words still have a search space of size 2^n . Write

$$D_{01}(n) = \max_{a \in \{0,1\}^n} |\det C(a)|.$$

The exact binary-circulant tables of Brent and Yedidia extend through order 53 in their updated work [1]. The order-54 computation ALETHEIA-54 provides a subsequent exact computational reference [2]. We determine both alphabet variants at order $55 = 5 \cdot 11$.

The novelty statement is intentionally scoped. The source and retrieval record used for this manuscript reports no prior published exact determination of $D_{01}(55)$ in the sources checked as of 9 September 2026; the search is not an absolute-priority claim. The checked records include the OEIS sequence and b-file, the Brent–Yedidia sources, ALETHEIA-54, and targeted searches for the exact integer and for order-55 binary circulants [3].

We obtain the following two headline results.

Theorem 1 (Global maximum and classification). *For binary circulant matrices of order 55,*

$$D_{01}(55) = 134694094094758395331307111329132.$$

Moreover, every maximizing word has weight 28. The maximizing words form one orbit under the affine action

$$a_i \mapsto a_{ui+t}, \quad u \in (\mathbb{Z}/55\mathbb{Z})^\times, \quad t \in \mathbb{Z}/55\mathbb{Z},$$

and this orbit has 2200 elements.

Theorem 2 (Sign maximum and classification). *For sign circulants of order 55,*

$$D_{\pm}^{\text{circ}}(55) = 297532404423965431213849494795385191907933028352,$$

$$S_{\pm}(55) = D_{\pm}^{\text{circ}}(55)/2^{54} = 16516366298178510386365752382353.$$

All maximizers have 23 or 32 plus signs, and row sum -9 or 9 . They form two affine orbits, each of size 2200 with trivial stabilizer, and one class under affine maps and global negation.

The canonical binary representative for Theorem 1 is

$$\boxed{0000000100011111011011001101011011010100011110101000111}$$

and has weight 28. It has stabilizer size one under the affine action. Its complement is

$$1111111011100000100100110010100100101011100001010111000,$$

of weight 27, and its determinant is

$$129883590734231309783760428781663.$$

The scope of Theorem 1 is binary circulant matrices, not all binary matrices and not unrestricted $\{\pm 1\}$ matrices.

The proof architecture is worth making explicit:

$$\begin{aligned} &\text{all } 2^{55} \text{ words} \longrightarrow \text{complement and weight reduction} \\ &\longrightarrow \text{exact spectral bounds} \longrightarrow \text{mod-5/mod-11 folds} \\ &\longrightarrow \text{correlation profiles} \longrightarrow \text{binary lifts} \\ &\longrightarrow \text{exact determinant comparison.} \end{aligned}$$

Each arrow is a mathematical reduction with an exhaustive finite residue, rather than a claim that an unstructured program happened to search far enough.

2 Binary circulants and Fourier formulation

Let $a \in \{0, 1\}^{55}$, let $k = \sum_{j=0}^{54} a_j$, and define

$$C(a)_{ij} = a_{j-i}, \quad f_a(x) = \sum_{j=0}^{54} a_j x^j.$$

If $\zeta = \exp(2\pi i/55)$, the Fourier eigenvalues of $C(a)$ are $f_a(\zeta^r)$, $0 \leq r < 55$, up to the harmless choice of frequency orientation. Since a is real and 55 is odd, the nonzero frequencies pair by complex conjugation. Thus, with

$$q_r = |f_a(\zeta^r)|^2, \quad 1 \leq r \leq 27,$$

we have

$$\det C(a) = k \prod_{r=1}^{27} q_r \geq 0. \tag{1}$$

Consequently the absolute value in the definition of $D_{01}(55)$ is redundant for actual binary words, although absolute products are retained when formal correlation profiles have not yet been shown realizable.

Define the periodic autocorrelation sequence

$$c_s = \sum_{t=0}^{54} a_t a_{t+s}, \quad s \in \mathbb{Z}/55\mathbb{Z}.$$

Then $c_0 = k$, $c_s = c_{55-s}$, and counting ordered pairs of distinct support elements gives

$$2 \sum_{s=1}^{27} c_s = k(k-1). \quad (2)$$

The correlation sequence is the inverse Fourier transform of the squared Fourier magnitudes. Parseval therefore gives the two moment identities

$$2 \sum_{r=1}^{27} q_r = 55k - k^2, \quad (3)$$

$$k^4 + 2 \sum_{r=1}^{27} q_r^2 = 55 \left(k^2 + 2 \sum_{s=1}^{27} c_s^2 \right). \quad (4)$$

The first identity and AM–GM immediately imply

$$\det C(a) \leq k \left(\frac{k(55-k)}{54} \right)^{27}. \quad (5)$$

3 Complement duality and affine symmetry

Let $\bar{a} = 1 - a$. At the zero frequency, $f_{\bar{a}}(1) = 55 - k$. At every nonzero 55th root of unity,

$$f_{\bar{a}}(\zeta^r) = -f_a(\zeta^r), \quad 1 \leq r < 55,$$

because the sum of all 55th roots in a nontrivial frequency is zero. There are 54 nonzero eigenvalues, so the signs cancel in the determinant. For $0 < k < 55$ this proves

$$k \det C(\bar{a}) = (55 - k) \det C(a). \quad (6)$$

The constant words are singular. If

$$D_{\text{circ}}(55, k) = \max_{\text{wt}(a)=k} \det C(a),$$

then complement is a bijection between weight strata and

$$D_{\text{circ}}(55, 55 - k) = \frac{55 - k}{k} D_{\text{circ}}(55, k). \quad (7)$$

It is therefore enough to handle $1 \leq k \leq 27$. For the complementary word, the first-moment bound takes the form

$$\det C(\bar{a}) = (55 - k) \prod_{r=1}^{27} q_r \leq (55 - k) \left(\frac{k(55 - k)}{54} \right)^{27}. \quad (8)$$

Translations and multipliers preserve the determinant. Translation changes Fourier eigenvalues only by phases whose product is one, while multiplication of indices by $u \in (\mathbb{Z}/55\mathbb{Z})^\times$ permutes the nonzero frequencies. The affine group has order

$$55\varphi(55) = 55 \cdot 40 = 2200.$$

Reversal is the multiplier $u = -1$. Orbit sizes must still be computed from actual stabilizers; the group order alone does not imply that every orbit has size 2200. Correlations are invariant under translation and are indexed by the 27 opposite-shift pairs. On these pairs the effective unit action has order 20.

4 Spectral-moment reduction

The correlation identities turn the global problem into a finite collection of integer moment problems. Among h nonnegative integers with fixed sum T , write $T = hb + r$, $0 \leq r < h$. Their minimum square sum is

$$(h - r)b^2 + r(b + 1)^2.$$

Indeed, transferring one unit from two entries that differ by at least two strictly decreases the square sum. Applying this to the 27 independent correlations, and combining it with the exact product bounds below, excludes all lower weights $k \leq 24$ at threshold M .

For clarity, the remaining necessary windows are listed here. The entries are for $\sum_{s=1}^{27} c_s^2$; the upper endpoint is the largest integer that can still reach M under the certified moment bound.

lower weight k	complement weight	minimum square sum	maximum square sum
25	30	3336	3338
26	29	3913	3919
27	28	4563	4570

For $k = 27$, the deviations $d_s = c_s - 13$ have sum zero, so $\sum d_s^2$ is even; the final row is equivalently a small deviation window around the nearly balanced correlation 13.

4.1 Exact product bounds

The fourth moment in (4) fixes the square sum of the q_r . The following elementary extremal calculation supplies a rigorous upper bound on their product. Consider h nonnegative variables with sum S and square sum at least L . Contracting variables toward their common mean increases the product until the square-sum constraint is tight (or the variance is zero). At a positive constrained stationary point, Lagrange multipliers give

$$\frac{1}{x} = \alpha + 2\beta x,$$

so there are at most two distinct values. If the smaller value occurs m times, put $\mu = S/h$ and $V = L - S^2/h$. The two values are

$$\mu - \sqrt{\frac{V(h-m)}{hm}}, \quad \mu + \sqrt{\frac{Vm}{h(h-m)}}. \quad (9)$$

Enumerating $m = 1, \dots, h-1$, together with the zero-variance and boundary cases, exhausts the compact feasible set. The implementation bounds the square roots outward by integer arithmetic at a decimal scale of 10^{50} . No floating-point comparison decides a pruning step.

The certificate also imposes a uniform spectral cap. Let

$$b = \left\lfloor \frac{k(k-1)}{54} \right\rfloor, \quad \delta_s = c_s - b.$$

For every nonzero frequency,

$$q_r = k - b + \sum_{s=1}^{27} \delta_s 2 \cos(2\pi r s / 55). \quad (10)$$

The rearrangement inequality pairs the sorted deviations with sorted cosine coefficients. There are only three coefficient multisets, represented by $r = 1, 5, 11$, according to the gcd class of the frequency. Their rational outward bounds give a cap B valid for every q_r .

To maximize a product subject also to $0 \leq x_i \leq B$, enumerate the number of coordinates on the upper boundary. The remaining interior coordinates again have the form (9); infeasible sums, negative roots, and roots above B are discarded. This is a finite description of all extrema on the capped set. The cosine intervals are constructed from Machin's identity and rational Taylor remainders for π and cosine, and the stored dyadic intervals are independently checked by the verifier.

5 The 5×11 cyclotomic decomposition

The factorization $55 = 5 \cdot 11$ splits the frequency space into the cyclotomic sectors of orders 1, 5, 11, 55. Their dimensions are

$$1, \quad 4, \quad 10, \quad 40, \quad 1 + 4 + 10 + 40 = 55.$$

The nonzero sectors have respectively 2, 5, and 20 conjugate pairs. Let

$$N_m(a) = \text{Res}(f_a, \Phi_m), \quad m \in \{5, 11, 55\}.$$

All three cyclotomic degrees are even, so the resultant convention used here has positive sign, and conjugate pairing makes these norms nonnegative. We obtain

$$\det C(a) = k N_5(a) N_{11}(a) N_{55}(a). \quad (11)$$

Use the Chinese remainder identification

$$\mathbb{Z}/55\mathbb{Z} \cong \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/11\mathbb{Z}.$$

For a word a , define the two fold vectors

$$r_i = \sum_{t \equiv i \pmod{5}} a_t, \quad s_j = \sum_{t \equiv j \pmod{11}} a_t.$$

Thus $0 \leq r_i \leq 11$ and $0 \leq s_j \leq 5$. The order-5 norm N_5 depends only on r , and N_{11} only on s . If

$$h_i^{(m)} = \sum_{t \equiv i \pmod{m}} c_t,$$

then $h^{(m)}$ is the cyclic correlation of the corresponding fold. The sector moments are

$$S_1^{(m)} = \frac{m \sum_i r_i^2 - k^2}{2}, \quad (12)$$

$$S_2^{(m)} = \frac{m \sum_i (h_i^{(m)})^2 - k^4}{2}. \quad (13)$$

For fixed folds r and s , the first moment in the primitive order-55 sector is

$$T = \frac{55k + k^2 - 5 \sum_i r_i^2 - 11 \sum_j s_j^2}{2}. \quad (14)$$

Its norm is at most $(T/20)^{20}$. A simpler necessary fold screen is also useful. If the selected small sector has $h = (m - 1)/2$ pairs and sum T_m , the remaining $27 - h$ pairs have sum

$$R = \frac{k(55 - k)}{2} - T_m.$$

Any word whose complementary determinant is at least M must satisfy

$$(55 - k)N_m \left(\frac{R}{27 - h} \right)^{27 - h} \geq M. \quad (15)$$

This is rational and exhaustive over the bounded integer fold vectors.

The retained fold representatives, up to their affine fold symmetries, are:

k	order-5 representatives	order-11 representatives
25	17	321
26	27	560
27	26	660

All unit orientations of the folded correlations are retained. Independent translations of the two fold vectors lift jointly by the Chinese remainder theorem. This reduction does not assert that every pair of fold profiles has a binary lift; that remaining existence question is handled exhaustively in Section 7.

6 Correlation profiles and exact enumeration

For each active weight $k \in \{25, 26, 27\}$, the exact sum in (2), the square-sum window, the range $0 \leq c_s \leq k$, and the capped moment bound define a finite family of integer correlation profiles. The generator first visits sorted multisets and uses the multinomial coefficient to recover the exact number of ordered profiles. It then tests every coordinate assignment against both fold signatures and the effective order-20 unit action.

The resulting certificate counts are shown below. “Allowed” is the recorded overapproximation of formal ordered profiles after the initial integer screen. “Explicitly enumerated” is completed work, not projected throughput. A formal profile is not treated as a word until it passes the binary lift.

k	allowed ordered profiles	explicitly enumerated	canonical profiles	lift targets
25	407,277	2,925	55	55
26	32,178,654	816,102	22,375	281
27	6,043,753	6,026,203	155,558	1,980
total	38,629,684	6,845,230	177,988	2,316

For a formal profile, the q_r obtained from the discrete Fourier transform need not all be nonnegative. The profile screen therefore uses the safe inequality

$$\left| (55 - k) \prod_{r=1}^{27} q_r \right| \leq (55 - k) \left(\frac{\sum_r q_r^2}{27} \right)^{27/2}, \quad (16)$$

which is just Cauchy–Schwarz followed by AM–GM on the absolute values. The signed product is reconstructed from two 61-bit Fourier fields. A separate squared integer comparison proves that the product of the primes exceeds the bound in (16); hence no modular wrap can turn a formal product into a false target. Profiles whose exact absolute product is below M cannot contain an improving word. Profiles at or above the threshold are passed to the complete lift enumeration.

7 Exhaustive binary lifts

Use the explicit torus coordinate

$$(i, j) \in \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/11\mathbb{Z} \mapsto 11i + 45j \pmod{55}.$$

The 11 columns are five-bit masks, so there are 32 possible column states. For every retained correlation target, all compatible oriented pairs of fold vectors are enumerated after normalizing the independent row and column translations. This produces exactly 16,084 lift jobs.

The production lift splits the eleven columns into five and six. A partial assignment records its word and a seven-coordinate signature: five row counts and the two within-column correlation sums for shifts 11 and 22. These statistics are additive across columns. The left half is sorted by its packed signature; each right half is joined to the unique complementary signature. The packing is injective over the bounded coordinate ranges, so the join uses exact integer equality rather than a hash collision. Every joined word has the requested folds and the two stored correlations; the remaining independent shifts are checked by rotated 55-bit intersections and population counts.

Nonnegative partial row counts and partial correlations above their targets are safe pruning rules. No floating-point operation is used in the lift. The production run tests 15,487,882,832 joined words and finds exactly two normalized solutions. The canonical verifier replays all jobs with a six-plus-five split and compares joined-word counts and solution sets. This was initially a split-dependent cross-check. In this joint revision, the separate typed-key implementation exhausts all 16,084 original binary tasks with split six-plus-five. All 15,487,882,832 joined words and both normalized solutions agree task by task with production. The complete supplement is `certificates/order55_sign/binary_companion/full_fiber_report.json`. A separate small fiber is checked against all 78,125 direct column assignments using three splits.

8 Winner and classification

For the canonical winner w , the exact folded data and cyclotomic norms are

```
weight = 28
mod-5 margin: [5,4,8,6,5]
mod-11 margin: [3,1,4,2,3,1,1,3,3,3,4]
N5 = 131
N11 = 434039
N55 = 84603917386870862636041
```

Equation (11) gives $28 \cdot 131 \cdot 434039 \cdot 84603917386870862636041 = M$. The complement has the same three nontrivial norms and the zero-frequency factor changes from 28 to 27, giving the determinant displayed after Theorem 1.

The nonzero correlations of the winner have the distribution

$$13 \text{ four times,} \quad 14 \text{ forty-six times,} \quad 15 \text{ four times.}$$

Among the independent shifts $1 \leq s \leq 27$, the exceptional shifts are 2, 6 with value 15 and 7, 15 with value 13. For the complement, the half-profile deviations from 13 have square sum 4. If the

word is converted to signs by $x_t = 2a_t - 1$, its nonzero periodic autocorrelations are $-5, -1, 3$ with total squared energy 182 over all 54 nonzero shifts.

For completeness, the perfectly balanced weight-27 possibility would have $c_s = 13$ for every nonzero shift, corresponding to a cyclic $(55, 27, 13)$ difference set. It is impossible. Indeed, with $\alpha = f_a(\zeta_5)$, where ζ_5 is a primitive fifth root, the ideal correlations imply $\alpha\bar{\alpha} = 14$. The polynomial Φ_5 is irreducible modulo 2, since $\text{ord}_5(2) = 4$; hence $\mathbb{Z}[\zeta_5]/(2)$ is a field. Reducing the norm equation modulo 2 forces $\alpha = 2\beta$. Then $\beta\bar{\beta} = 7/2$, impossible because a rational algebraic integer is an integer. This obstruction is also discussed in the design literature [4]. The actual winner is not being presented as a new difference set or as a two-level almost difference set; the exact classification comes from the exhaustive lift results.

The global classification now follows. Any word with determinant at least M has a lower complementary weight $1 \leq k \leq 27$. The exact weight screen excludes $k \leq 24$. For $k = 25, 26, 27$, its moment identities place the correlation multiset in the enumerated finite family; the folded screens place both fold vectors among the retained representatives; and a unit transformation takes the paired correlations to one of the 2,316 targets. The production lift certificate contains a transformed copy of the word. The two normalized lifted solutions both give complementary determinant M and have the same affine canonical representative. The newly completed independent full-fiber computation discharges the remaining union condition in the historical review snapshot.

Finally, direct enumeration of the affine action on the canonical word gives 2200 distinct words. Its stabilizer therefore has size one, and the exhaustive lift classification shows that all equality cases lie in this one orbit. This proves Theorem 1.

9 Computer-assisted certification

The proof separates discovery from certification.

Heuristic stage. Random or local search was used only to find a strong incumbent. Search scores, partial jobs, and floating-point rankings are not evidence of maximality.

Rigorous stage. Exact first- and fourth-moment bounds, rational outward intervals, exhaustive folded-profile coverage, exhaustive binary lifts, exact integer arithmetic, and deterministic certificate checks prove the global statement.

The final winner is checked through three mathematical routes:

1. finite-field Fourier determinants reconstructed by signed CRT;
2. an exact integer matrix determinant using fraction-free elimination;
3. the product of the three exact cyclotomic resultants.

The production verifier does not import the search or certificate-builder module. It re-derives the rational bounds, checks the Machin/Taylor cosine enclosures, verifies retained fold norms by resultants, recompiles the hash-identified native generators, regenerates the correlation profiles, and checks task identities, multiplicities, exact output words, determinants, and the affine classification. Its optional full mode is a split-dependent six-plus-five replay of the native lift algorithm. The final global certificate records the component SHA-256 hashes and the counts used in this paper; the independent completeness audit is a separate, explicitly qualified review layer.

10 Completeness statements and review status

The global argument uses three logically distinct finite statements. We state them separately because reproducibility of a computation is not the same as an independently checkable certificate of its coverage.

Lemma 1 (Correlation-spectrum coverage). *For each active weight $k \in \{25, 26, 27\}$, the integer moment equations, the fold screens, and the unit action reduce every necessary correlation profile to one of the finite profile records used in the certificate.*

For unambiguous machine-readable transcription, the corresponding certificate integers are 2316 correlation targets, 6845230 explicit profile evaluations, 16084 lift tasks, and 15487882832 joined words.

The clean-room profile evaluator `audit/completeness/run_cleanroom_profile_audit_v4.py` reconstructs the three ordered-profile families without importing the production verifier or calling a production executable. It reports exact agreement of the 16 profile-family set, exact agreement of the 2,316 target triples, and 982,589 canonical profile records below the incumbent. This is the independently checkable portion of the profile-spectrum reduction.

Lemma 2 (Pruning soundness). *Every profile recorded below the incumbent in the exact profile-product stage is rigorously pruned for the purpose of reaching M ; every retained profile is represented by a lift target and is not pruned by that exact profile-product comparison.*

The clean-room records make this statement inspectable as a machine-readable table with fields `profile_id`, `profile_data`, `bound_type`, `exact_upper_bound`, `incumbent`, and `pruned`. The separate rational boxed-moment rows remain part of the production bound certificate and are checked by the canonical verifier.

Lemma 3 (Lift completeness). *Every binary word with determinant at least M has a transformed representative in the union of the 16,084 listed margin fibers, each of which is exhausted by production and independent implementations. The preceding profile and pruning lemmas therefore imply Theorem 1.*

Proof. The target and task sets were independently reconstructed in the earlier review. The new supplement exhausts their entire fibers with the separate typed-key implementation and checks exact task counts and word sets. CRT combines the two fold rotations into a translation modulo 55, and unit normalization preserves the determinant. Thus the listed fibers cover a representative of every eligible word. $\square$

The historical v2 independent lift report is preserved unchanged. Its previously missing full-fiber step is now supplied by `certificates/order55_sign/binary_companion/full_fiber_report.json`. The binary value and frozen original certificates are unchanged.

11 Reproducibility and availability

The production theorem is tied to commit `e77a1aa6afea16fe59a35f9c0de93f385f9b8781`. The repository contains the exact certificate directory, the native source files, and the verifier. From the repository root, the fast publication checks are:

```
python scripts/verify_order55_global.py
python -m pytest
python -m pip check
```

The split-dependent six-plus-five lift replay is requested with

```
python scripts/verify_order55_global.py --full-lifts
```

The native replay requires a C++20 compiler with unsigned `__int128` support. The stored build commands and source hashes identify the exact native inputs; host-specific paths and elapsed times may differ, but the discrete counts, words, products, and classification must agree.

The project repository is <https://github.com/dongxuelian2/order55-binary-circulant-maxdet>. The joint manuscript and OEIS text are included in this publication release. The versioned PDF is attached to the project release. The stored sign proof is checked by `pythonscripts/verify_order55_sign.py`; a full independent replay can be requested with the `--replay` option.

12 The companion sign-circulant maximum

Write $D_{\pm}^{\text{circ}}(55)$ for the maximum absolute determinant over sign circulants and $S_{\pm}(55) = D_{\pm}^{\text{circ}}(55)/2^{54}$. These are OEIS A215723 and A215897 [5, 6]. All sign screening is regenerated; the old binary target set is not a completeness certificate for this different objective.

12.1 Exact bridge and complement reduction

Lemma 4 (Binary-to-sign bridge). *For $a \in \{0, 1\}^{55}$ of weight $0 < k < 55$ and $x = 2a - \mathbf{1}$,*

$$\det C(x) = 2^{54} \frac{2k - 55}{k} \det C(a), \quad S_{\pm}(55) = \max_{1 \leq k \leq 27} \frac{55 - 2k}{k} D_{\text{circ}}(55, k).$$

Proof. The identity $C(x) = 2C(a) - J$ changes the eigenvalue on the all-ones line to $2k - 55$ and doubles every nontrivial Fourier eigenvalue. It also holds when the circulant is singular. Conjugate pairing gives $\det C(a) = k \prod q_r$. Complementing a negates x and preserves absolute determinant. The constant endpoints have rank one. $\square$

Thus the reduced objective is

$$R(a) = (55 - 2k) \prod_{r=1}^{27} q_r = (55 - 2k) N_5 N_{11} N_{55}.$$

For the encoding $x = 2a - \mathbf{1}$, the signed row sum is $2k - 55$; $55 - 2k$ is its magnitude on the reduced weights. In contrast, the complementary binary objective has coefficient $55 - k$. The small-weight certificates in Appendix A transfer exactly by the bridge. The binary global theorem and complement duality further give the exact weight-27 sign value $M/28 = 4810503360527085547546682547469$. The sign upper-bound proof below does not require the binary global theorem.

12.2 Sign correlations and exact bounds

Put $\rho_s = \sum_i x_i x_{i+s}$. Expansion and orthogonality give

$$\rho_s = 55 - 4k + 4c_s, \quad \rho_s \equiv 3 \pmod{4}, \quad 55 + 2 \sum_{s=1}^{27} \rho_s = (2k - 55)^2.$$

For $p_r = |f_x(\zeta^r)|^2 = 4q_r$ and $\sigma = 2k - 55$,

$$\sigma^2 + 2 \sum p_r = 55^2, \quad \sigma^4 + 2 \sum p_r^2 = 55 \left(55^2 + 2 \sum_{s=1}^{27} \rho_s^2 \right).$$

At fixed weight this is an invertible affine change of the integer correlations; the sign congruence adds no condition beyond that integrality. The implementation keeps the sparse binary deviations.

The first-moment bound is

$$R(a) \leq (55 - 2k) \left(\frac{k(55 - k)}{54} \right)^{27}.$$

The fourth-moment and capped-product lemmas above apply to $\prod q_r$, with every threshold multiplied by $55 - 2k$. All exclusions are strict integer or rational outward comparisons against the exact incumbent. For a mod-5/mod-11 fold pair (r, s) , put

$$T_{55} = \frac{55k + k^2 - 5 \sum r_i^2 - 11 \sum s_i^2}{2}.$$

The joint-fold bound is

$$(55 - 2k)N_5(r)N_{11}(s)(T_{55}/20)^{20}.$$

The primitive fourth moment subtracts the two folded fourth moments from the full fourth moment, so the same product lemma can be used in that sector. The recorded joint screen combines the first-sector bound with the global fourth-moment bound; all conservative survivors are subsequently resolved by complete profiles and lifts.

12.3 Complete screening and independent audit

The exact moment screen leaves $k = 22, 23, 24, 25, 26$. Capped moments reduce 9,531,797,706 ordered formal correlations to 731,378,844 explicit evaluations in 17 multisets.

k	Formal profiles	Explicit profiles	Targets	Lift tasks
22	17,383,860	17,383,860	0	0
23	3,580,859,880	259,224,030	1,705	10,580
24	5,501,039,895	435,495,645	10,647	77,862
25	432,505,269	19,266,507	5,477	63,300
26	8,802	8,802	6	30

There are 17,835 canonical targets and 151,772 lift tasks. No target remains at weight 22.

The independent arithmetic checker parametrizes the two stationary values by their separation, verifies its own dyadic cosine enclosures, and generates multisets by descending multiplicities. An independent bounded-alphabet FKM necklace generator produces every cyclic fold representative; it shares no production affine-minimum test. All fold representative sets and their correlations agree exactly.

An independent `next_permutation` traversal computes every canonical algebraic product by CRT without production interval-product pruning. Its raw above-threshold set contains 255 additional indefinite formal spectra. For every one, a recorded frequency and exact rational upper bound prove $q_j < 0$, excluding a binary autocorrelation. After these explicit positive-semidefinite discharges, the independent and production target sets, including their products, are exactly equal.

The production five-plus-six lift uses a sorted packed integer key. The independent six-plus-five implementation uses typed seven-coordinate hash keys with equality checks and tests every shift.

12.6 Public sources and reproducibility

The check on 11 September 2026 inspected both OEIS entries, the JIS article and the latest Brent–Yedidia revision. The OEIS linked tables are described as ending at 52, whereas arXiv v6 Table 4 gives sign values through 53. Searches of the exact new integers, both encodings, arXiv, GitHub, Zenodo and the general web located no earlier exact sign-55 determination. To the best of our knowledge this is the missing companion result; the search does not establish absolute priority.

All sign certificates are under `certificates/order55_sign/`. The ordinary verifier rechecks hashes, exact winner arithmetic, weight bounds, target sets, negative-spectrum certificates, task records and orbit classification. The optional second command recompiles and independently reruns the complete proof without any heuristic search:

```
python scripts/verify_order55_sign.py
python scripts/verify_order55_sign.py --replay
```

13 Conclusion

We have determined the exact maximum determinants of 55×55 circulants over both alphabets and classified every equality case. The essential global step is the exhaustive chain from weight reduction to folded correlation profiles and then to binary torus lifts. The factorization $55 = 5 \cdot 11$ provides the cyclotomic sectors that make this chain finite and auditable, while the final exact arithmetic removes any dependence on numerical rounding. The resulting value is

$$D_{01}(55) = 134694094094758395331307111329132,$$

with one affine class of 2200 maximizers. The companion sign maximum is

$$16516366298178510386365752382353 \cdot 2^{54},$$

with 4400 words in two affine orbits paired by negation.

A Previously certified small-weight strata

The starting repository also contains an independent exact certificate for $D_{\text{circ}}(55, k)$ with $1 \leq k \leq 7$. These values are not a separate headline result; they are supporting fixed-weight strata and can be paired with their complements using (7).

k	$D_{\text{circ}}(55, k)$
1	1
2	2048
3	1564031349
4	6424318816256
5	12556202952818275
6	4342461217049330166
7	570999857161750276477

References

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